

# Peter (Pedram) Beigi, PhD Candidate

Autonomous Driving Researcher and Machine Learning Engineer

Based in Santa Clara, CA & Washington, DC  
Open to Relocate

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## PROFESSIONAL SUMMARY

Develops perception, prediction, and decision-making systems for autonomous driving. Experienced in building scalable ML pipelines using large-scale multimodal datasets across simulation and real-world environments.

## RESEARCH EXPERIENCE

### Research Intern

Exp: Sep 2026 - Dec 2026

 University of Illinois Urbana-Champaign

Urbana-Champaign, IL

- Will work with [Prof. Talebpour](#) on developing advanced approaches including foundation models, vision-language architectures, diffusion models, imitation learning, and reinforcement learning, leveraging large-scale driving data.

### Autonomous Vehicle & Machine Learning Researcher

Sep 2022 - Present

 FHWA Turner-Fairbank Research Center (Saxton Lab)

McLean, VA

- Leading development of trajectory generation models for ADAS (using physics-based, DL-based and RL-based frameworks), and integrating AV behavior models into large-scale microsimulation (SUMO) tools (*Repos are private for now*).
- Developed and validated perception (sensor fusion, detection, segmentation) and planning (map generalization, localization) algorithms using LiDAR, radar, and video data for enhanced AV situational awareness and mixed-traffic simulation (*Some repos are public here*).
- Built an end-to-end computer vision pipeline (Faster R-CNN, multi-camera MOT, BEV transformation, recursive state estimation, SIFT feature matching, and annotation/data processing) to extract trajectories and support motion prediction on 11,000+ vehicle-km of AV-human interaction data (*Data is published by USDOT*).

### Graduate Researcher

Sep 2021 - Present

 The George Washington University

Washington, DC

- Developed multi-agent reinforcement learning (MARL) frameworks for AV coordination and negotiation in dynamic partially observable environments using CTDE framework with multi-agent PPO actor-critic models. Applied Bayesian optimization and genetic algorithms for the models' calibration.
- Engineered model observability and traceability by developing Logsim DAGs to capture closed-loop metrics, improving experiment reproducibility for the autonomous stack.
- Led creation of an integrated simulation-AI pipeline combining SUMO, CARLA, and Python APIs for closed-loop decision-making experiments using real-time data (trajectory) streaming using twelve 5G-connected cameras.

## WORK EXPERIENCE

### Autonomous Vehicle Researcher Intern

Sep 2024 - Dec 2024

 moment.ai

Washington, DC

- Developed multi-modal deep learning time-series models (LSTM/GRU) integrating smartphone and AV sensor data to detect driver behavior and health incident risks, outperforming traditional methods by 9%.
- Developed a real-time NVIDIA Jetson/mobile driver monitoring system using AWS IoT pipelines and Alexa integration for in-vehicle alerts and control.

### Data Analyst Intern

May 2025 - Aug 2025

 Mead & Hunt

Washington, DC & Columbia, MD

### Data Analyst Intern

May 2024 - Aug 2024

 Arcadis

Atlanta, GA

## SELECTED PUBLICATIONS

**h-index: 6 / citation: 219**

|                                                                                                                                    |                          |
|------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| "Understanding the Interaction of Vehicles within an Intersection: An Optimal Control Approach with Differential Games"            | INFORMS (2026)           |
| "Diffusion Process-Based Model for Network Trajectory Propagation"                                                                 | IEEE T-ITS (2026)        |
| "Percolation-Based Cloud Computing Framework for Network Reliability Assessment"                                                   | IEEE ICMI (2025)         |
| "Learning to Negotiate Chaos: Residual Multi-Agent Policies over Behavioral Priors"                                                | IEEE ICRA (under review) |
| "Deep Reinforcement Learning Under Partial Observability: Co-Simulation and Penetration-Aware Policy Learning for Traffic Control" | IEEE VTC Fall (2026)     |
| "A Hybrid Genetic Algorithm and Reinforcement Learning Approach for Vulnerable Road User Modeling"                                 | TRR (under review)       |
| "A Deep Reinforcement Learning Control to Explore Impact of V2X Connectivity on Intersection Performance"                          | TRR (2025)               |
| "Third Generation Simulation Dataset"                                                                                              | TRR (2024)               |

## TECHNICAL SKILLS

**Programming:** Python, ROS, MySQL, Git

**Libraries:** PyTorch, Diffusers, Transformers, OpenCV, RLib, Gymnasium, Scikit-learn, Keras, Ultralytics

**Cloud Computing Platform:** Azure ML

**Datasets:** Waymo Motion, Waymo Perception, nuScenes, Argoverse 2, TGSIM, HighD, INTERACTION

**Tools:** CARLA, Unity, Power BI, Tableau, SUMO, VISSIM, Synchro, SIDRA

## EDUCATION

### The George Washington University

Sep 2021 - Exp: May 2027

PhD in Transportation Engineering (AV Systems & AI) - GPA: 3.91/4.00

### Sharif University of Technology

Sep 2016 - Feb 2021

Bachelor of Science in Civil Engineering - GPA: 3.80/4.00

## PROFESSIONAL CERTIFICATIONS

Social and Behavioral Research Certification

Issued by: Collaborative Institutional Training Initiative (CITI Program)

Credential ID: 71300833 | Aug 2021 - Aug 2027

## RELEVANT COURSEWORK

- UMD: CMSC 422 Machine Learning (audited)
- UMD: CMSC 472 Deep Learning (audited)
- GW: MAE 6292 Robotics Vision and Perception (4/4)
- GW: SEAS 6520 Autonomous Systems and Robotics (audited)
- Stanford: CS231n DL for Computer Vision (Online)
- GW: CE 6609 Numerical Methods in Engineering (4/4)
- GW: CE 6800 Data Science (4/4)
- GW: ECE 6130 Big Data and Cloud Computing (4/4)
- UMD: CMSC 426 Computer Vision (audited)
- UMD: CMSC 828 Visual Learning and Recognition (audited)
- Google DeepMind: Introduction to RL (Online)

## WORK AUTHORIZATION

No sponsorship needed.